

URL Shortener

Functional Requirements-

- System should be able to accept a long URL and output a short URL.
- The short URL the system outputs should redirect the user to the original long URL.
- The system should validate all URL's and reject all invalid URL's.
- The system must ensure that all short URL's generated are unique.
- The generated URL's must be as short as possible.

Non Functional Requirements-

- Redirect requests should have low latency.
- Handle high volumes of concurrent redirect requests.
- Use HTTPS for communication.
- Separate Shortener and Redirect services.
- Once a short URL is created, it should consistently resolve to the same original URL.

Database Structure

URL's	
PK	<u>UniqueID : BIGINT</u>
UQ	short_code : VARCHAR(10) original_url : TEXT created_at : TIMESTAMP expires_at : TIMESTAMP click_count : BIGINT is_active : BOOLEAN

API METHODS

1. Create Short URL

Method:
POST

Endpoint:
/shorten

Description:
Creates a new shortened URL.

Request:

```
{  
  "originalUrl": "https://www.example.com/articles/system-design/url-shortener",  
  "customAlias": "design101",  
  "expiresAt": "2026-12-31T23:59:59Z"  
}
```

Response (201 Created):

```
{  
  "id": 1024,  
  "shortCode": "design101",  
  "shortUrl": "https://short.ly/design101",  
  "originalUrl": "https://www.example.com/articles/system-design/url-shortener",  
  "createdAt": "2026-07-31T10:30:15Z",  
  "expiresAt": "2026-12-31T23:59:59Z"  
}
```

2. Redirect to Original URL

Method:

GET

Endpoint:

/{"shortCode"}

Description:

Retrieves the original URL and redirects the user.

Example:

GET /design101

Response:

HTTP/1.1 302 Found

Location: https://www.example.com/articles/system-design/url-shortener

Error (404):

```
{  
  "error": "Short URL not found."  
}
```

3. Retrieve URL Details

Method:

GET

Endpoint:

/url/{shortCode}

Description:

Returns metadata for the shortened URL.

Example:
GET /url/design101

Response:

```
{  
  "id": 1024,  
  "shortCode": "design101",  
  "originalUrl": "https://www.example.com/articles/system-design/url-shortener",  
  "clickCount": 154,  
  "createdAt": "2026-07-31T10:30:15Z",  
  "expiresAt": "2026-12-31T23:59:59Z",  
  "isActive": true  
}
```

4. Update Short URL

Method:
PUT

Endpoint:
/url/{shortCode}

Description:
Updates the destination URL or expiration date.

Request:

```
{  
  "originalUrl": "https://www.example.com/new-page",  
  "expiresAt": "2027-01-01T00:00:00Z"  
}
```

Response:

```
{  
  "message": "Short URL updated successfully."  
}
```

5. Delete Short URL

Method:
DELETE

Endpoint:
/url/{shortCode}

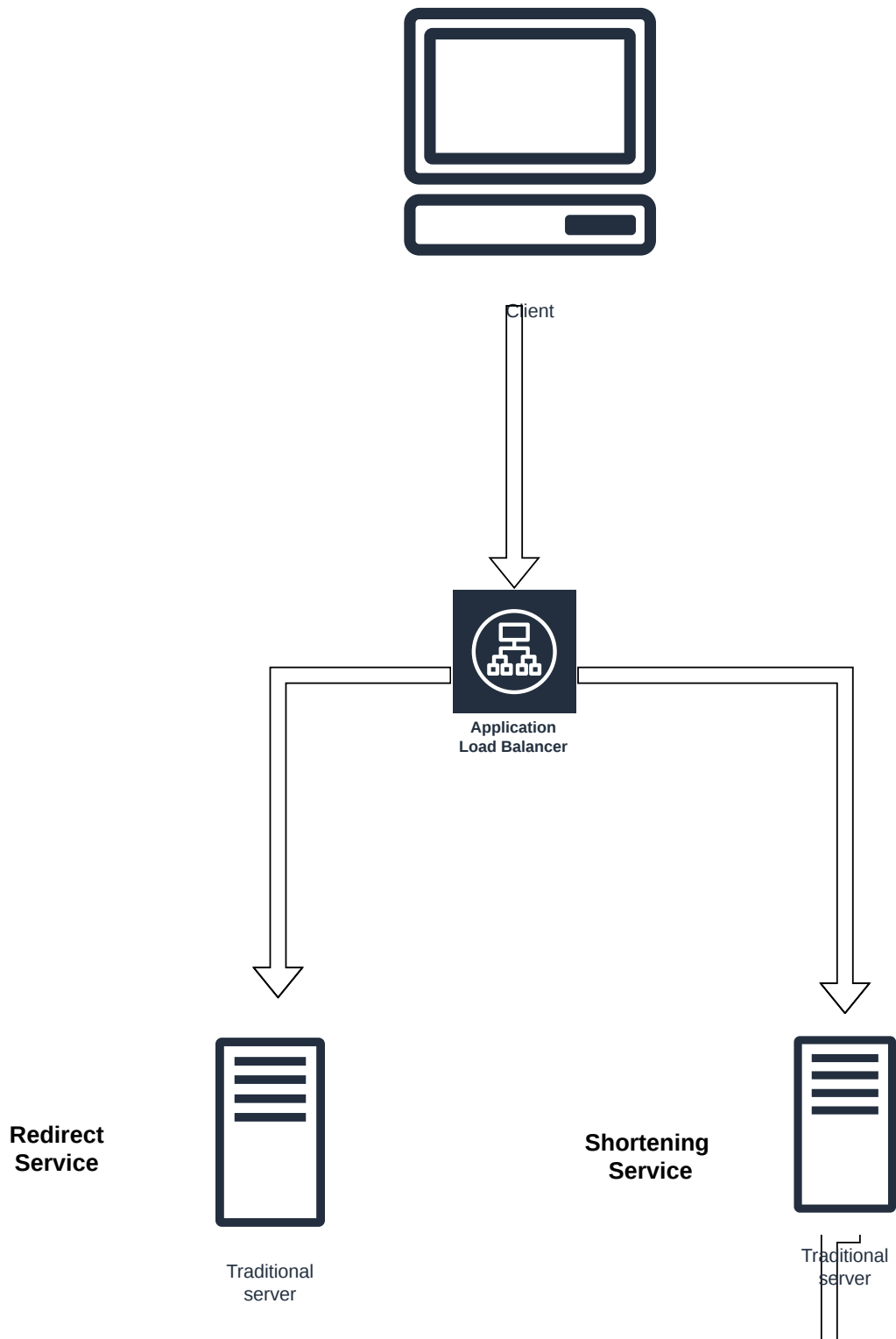
Description:
Deletes or deactivates a shortened URL.

Example:
DELETE /url/design101

Response:

```
{  
  "message": "Short URL deleted successfully."  
}
```

Architecture



Cache

Database

